



IBM Developer
SKILLS NETWORK

Winning Space Race with Data Science

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Outline

- Executive Summary
- Introduction
- Methodology
- Results
- Conclusion
- Appendix

Executive Summary

- **Summary of methodologies**

- Data Collection
- Data Wrangling
- Exploratory Data Analysis with Data Visualization(EDA)
- EDA with SQL
- Interactive Map with Folium
- Dashboard with Plotly
- Predictive Analysis

- **Summary of all Results**

- Exploratory Data Analysis Results
- Interactive Analytics Demo (screenshots)
- Predictive Analysis Results

Introduction

- **Project background and context**

SpaceX is a private aerospace company that has revolutionize space technology, by making space travel affordable. The company advertises Falcon 9 rocket launches on its website, with a cost of \$62m; other providers cost upward of \$165m each , much of the savings is because SpaceX can reuse the first stage. Therefore, if we can determine if the first stage will land , we can determine the cost of a launch. For this project, we shall examine several machine learning models and then predict if SpaceX will reuse the first stage.

- **Questions to answer**

What is the best machine learning algorithm that can be used for binary classification in this case?

Doest the rate of successful landings increase over the years?

Is there a relationship between variables such as payload mass, launch site, number of flights , orbits and the success of the first stage landing?

Section 1

Methodology

Methodology

- Data collection methodology:
 - Using SpaceX Rest API
 - Using Web Scrapping from Wikipedia
- Perform data wrangling
 - Filtering the data
 - Dealing with missing values
 - Using One-hot encoding to convert categorical variable to numerical for easy model training.
- Perform exploratory data analysis (EDA) using visualization and SQL
- Perform interactive visual analytics using Folium and Plotly Dash
- Perform predictive analysis using classification models
 - Build, tune, evaluate classification models to ensure best results

Data Collection

- Data collection involved API requests from SpaceX and Web Scraping from tables in SpaceX's Wikipedia website.
- Data Columns are obtained by using SpaceX REST API.
- Data columns were : Flight No. , Launch site, Payload Mass , Orbit, among others

Data Collection – SpaceX API

- Requesting rocket launch data from SpaceX API.
- Decoding the response content using `.json()` and turning it into a dataframe using `.json_normalize()`.
- Requesting needed info about the launches from SpaceX API by applying custom functions.
- Constructing data we have obtained into a dictionary.
- Creating a dataframe from the dictionary.
- Filtering the dataframe to only include Falcon9 Launches , remove Falcon 1.
- Replace missing values of Payload Mass column with calculated mean for this column.
- Exported the data to CSV.

https://github.com/krisotoo/Capstone-2026-C/blob/main/jupyter-labs-spacex-data-collection-api_Lab1_OTOO.ipynb

Data Collection - Scraping

- Request falcon 9 launch data from Wikipedia
- Creating a BeautifulSoup object from the HTML response.
- Extracting all column names from the HTML table header.
- Collecting the data by parsing the HTML tables
- Constructing data we have obtained into a dictionary.
- Creating a data frame from the dictionary.
- Exporting the data to CSV.

- https://github.com/krisotoo/Capstone-2026-C/blob/main/jupyter-labs-webscraping%20_Lab2_OTOO.ipynb

Data Wrangling

- Identified and replaced missing values with mean eg. avg_payload.
- Used value_counts() on column LaunchSite for number of launches in each site.
- Created a landing outcome label from Outcome column.
- Landing_class = 0 if bad outcome
- Landing_class – 1 otherwise.

https://github.com/krisotoo/Capstone-2026-C/blob/main/labs-jupyter-spacex-Data%20wrangling_Lab3_OTOO.ipynb

EDA with Data Visualization

✓ Some charts plotted:

☐ Flight Number vs Payload mass

☐ Flight Number vs Launch site

☐ Payload Mass vs Launch Site.

☐ Orbit Type vs Success Rate

☐ Flight Number vs Orbit Type

☐ Line charts and Bar charts

☐ The goal is to show if there is/are any significant relationships between these variables since they can be important for the machine learning model.

☐ https://github.com/krisotoo/Capstone-2026-C/blob/main/edadataviz_OTOO.ipynb

EDA with SQL

✓ Performed SQL queries:

- ☐ Display the names of unique launch sites.
- ☐ Display 5 records where launch sites begin with sting 'CCA'.
- ☐ Display the total payload mass carried by booster launched by NASA(CRS).
- ☐ Display the average payload mass carried by booster version F9 v1.1
- ☐ List the names of boosters which have success in drone ship and have payload mass greater than 4000 but less than 6000.
- ☐ List the total number of successful and failed mission outcomes.
- ☐ List the names of the booster versions which have carried the maximum payload mass.
- ☐ List the failed landing outcomes in drone ship, their booster versions and launch site names for months in year 2015
- ☐ Rank the count of landing outcome(such as Failure(drone ship) or Success(ground pad) between the date 2010-06-04 and 2017-03-20 in descending order

https://github.com/krisotoo/Capstone-2026-C/blob/main/jupyter-labs-eda-sql-coursera_sqllite%20_OTOO.ipynb

Build an Interactive Map with Folium

❖ Markers on all launch sites;

- Added marker with Circle, Pop-up label and Text label of NASA Johnson Space Center using its longitude and latitude coordinates as a start location. Added folium markers with circle, pop-up and text label of all launch sites using their longitude and latitude coordinates to show the geographical location and proximity to the equator and coast.

❖ Colored Markers of the launch outcomes for each launch site;

- Added colored markers(**green**) and failed(**red**) launches using marker cluster to identify which launch sites have relatively high success rates.
- https://github.com/krisotoo/Capstone-2026-C/blob/main/lab_jupyter_launch_site_location_OTOO.ipynb

Build a Dashboard with Plotly Dash

- Site selection dropdown: Options: "All Sites" + individual launch site
- Pie Chart(Success Analysis): Shows total success vs failure for specific sites.
- Slider of Payload Mass Range : Added a slicer to select Payload range.
- Scatterplot(Payload vs Success Correlation): color-coded by Booster version category
- Added a scatter chart to show flight number vs Launch Site: color-coded by Mission outcome(Success/Failure). I added this to show launch progression, launch site usage and success rate .

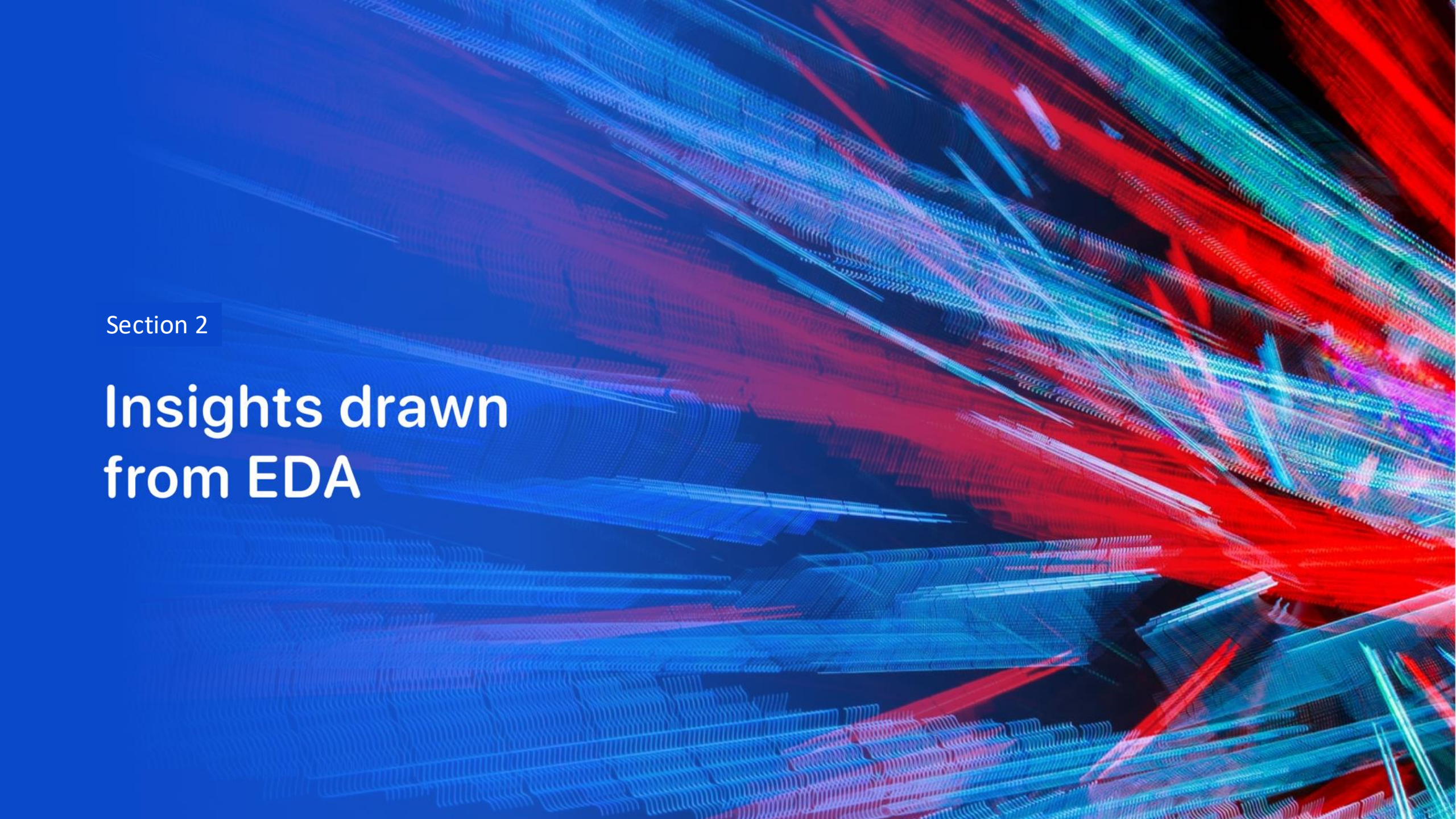
<https://github.com/krisotoo/Capstone-2026-C/blob/main/spacex-dash-app.py>

Predictive Analysis (Classification)

- Create a NumPy array from column “Class” in data and assigned to variable Y.
- Standardize data with `StandardScaler()` in X , reassign to X using `fit_transform` to transform the data.
- Use the function `train_test_split()` to split data into training and test sets 80% to 20% respectively.
- Create a GridsearchCV object with cv=10 to find best parameters.
- Apply GridsearchCV on `logreg`, `SVM`, `Decision tree` and `Knn models`.
- Calculated the accuracy on the test data using the method score.
- Examine the confusion matrix for all models .
- Compare and find method that performs best.
- https://github.com/krisotoo/Capstone-2026-C/blob/main/SpaceX_Machine%20Learning%20Prediction_Part_5_OTOO.ipynb

Results

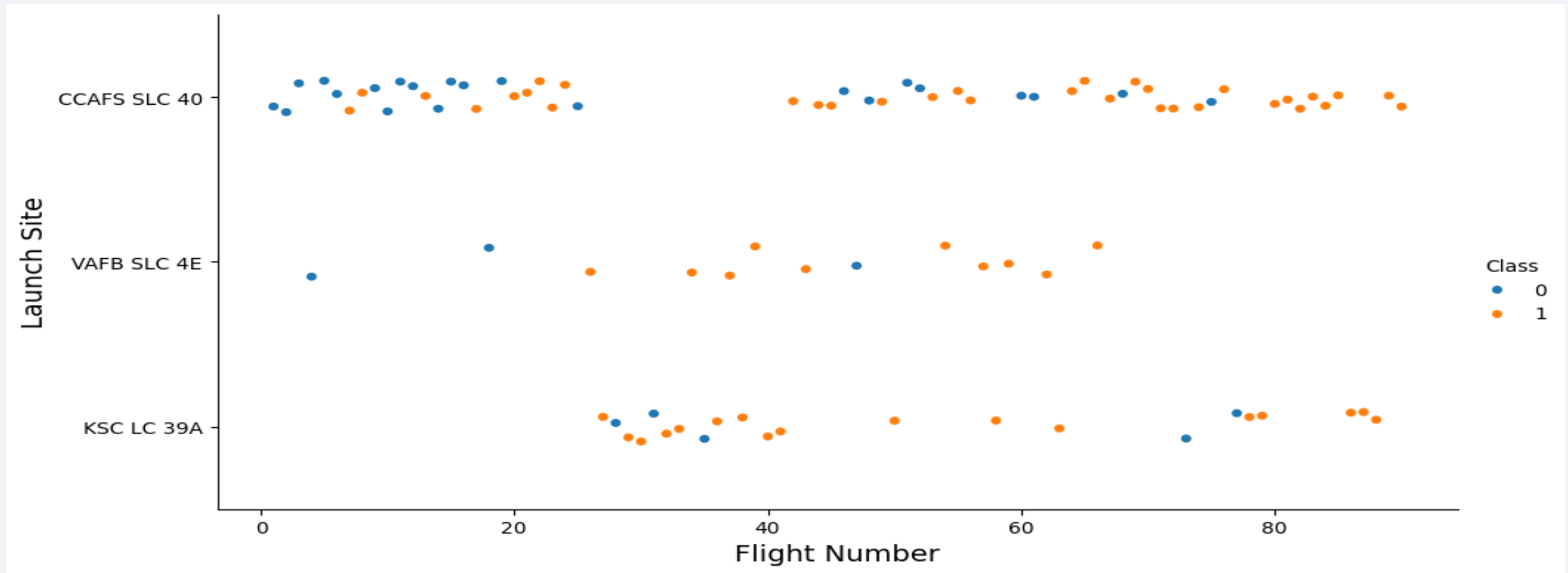
- Exploratory data analysis results
- Interactive analytics demo in screenshots
- Predictive analysis results

The background of the slide is an abstract composition. It features a dark blue field on the left side, which transitions into a complex pattern of diagonal streaks in shades of blue, red, and teal on the right. These streaks have a textured, almost woven appearance. Overlaid on this pattern is a faint, light blue grid that recedes into the distance, creating a sense of depth and perspective.

Section 2

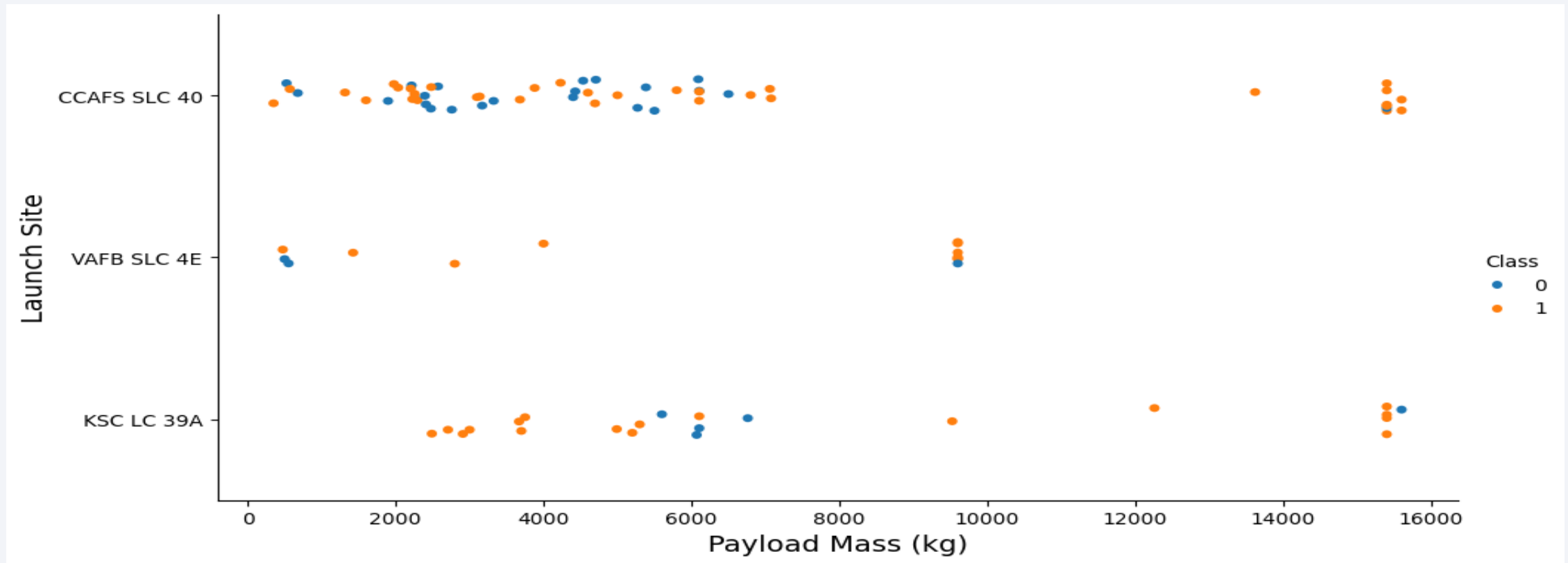
Insights drawn from EDA

Flight Number vs. Launch Site



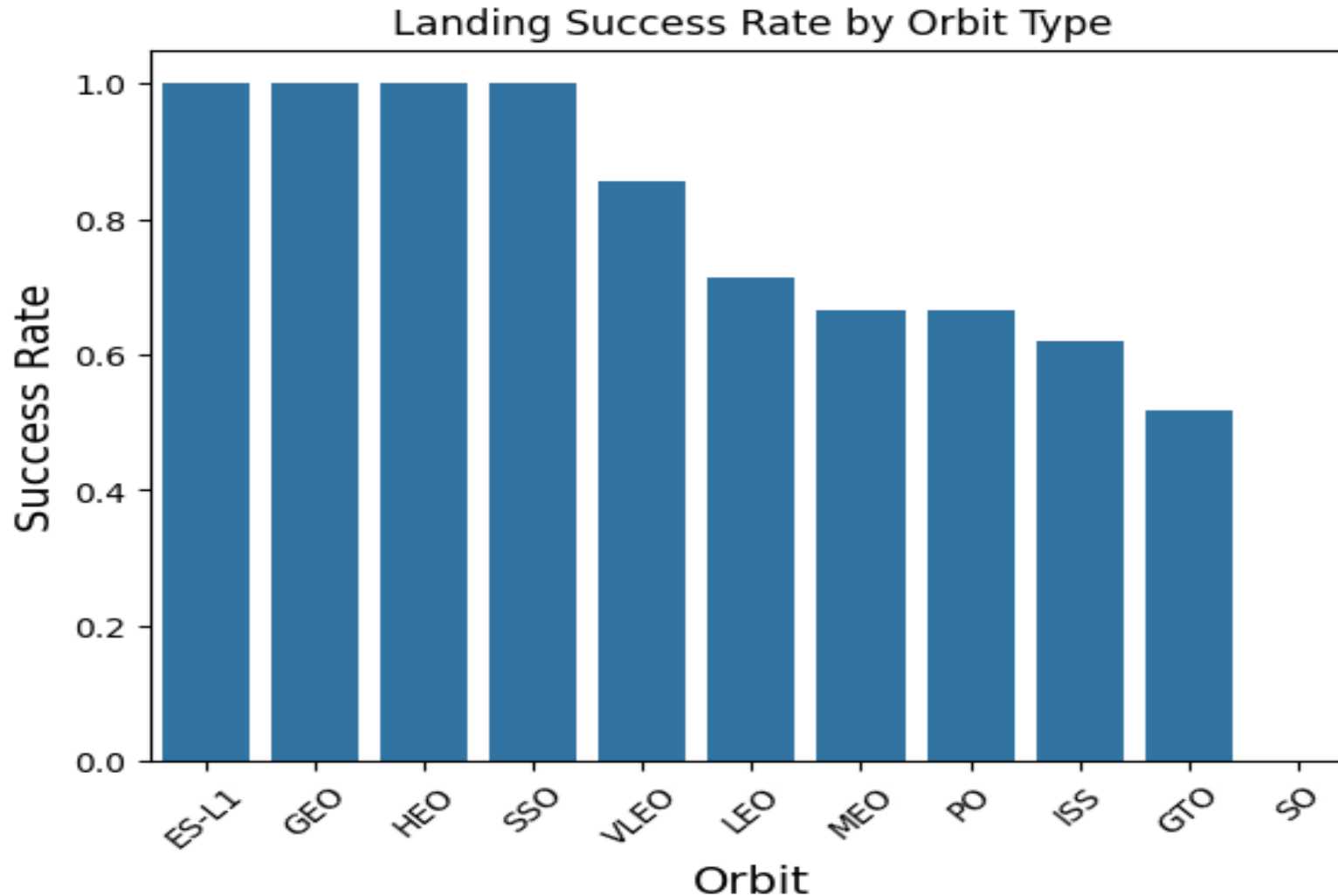
The plot shows that **success rate** improved significantly over time, with early flights having more **failures** and later flights being predominantly successful. It also shows the evolution of launch site usage, with **CCAFS SLC 40** being heavily used early on, while **KSC LC 39A** became more prominent in later, higher-numbered flights.

Payload vs. Launch Site



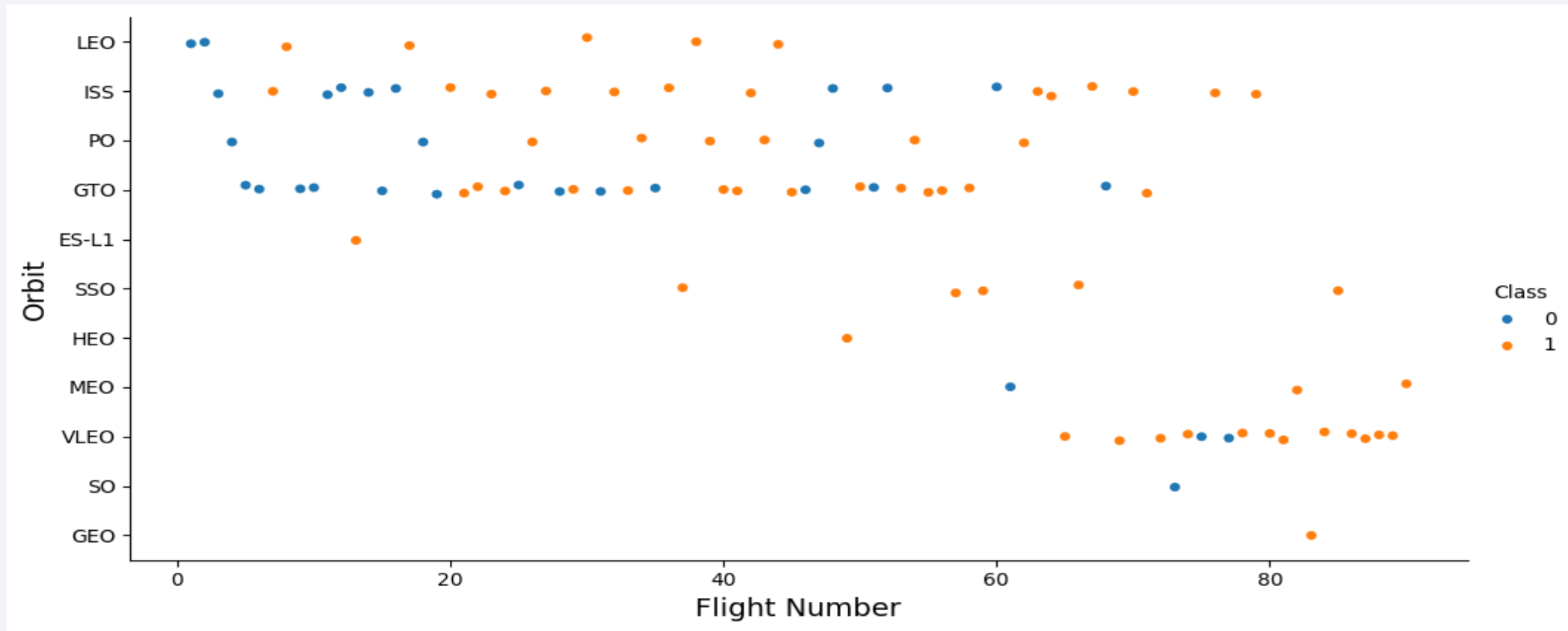
This scatter plot shows how payload mass varies by launch site, with mission success (orange) and failure (blue) indicated by color. Notably, the KSC LC 39A site has a 100% success rate for payload mass under 5500kg. Also, at every launch site, as payload mass increases, there is a higher chance of successful landing.

Success Rate vs. Orbit Type



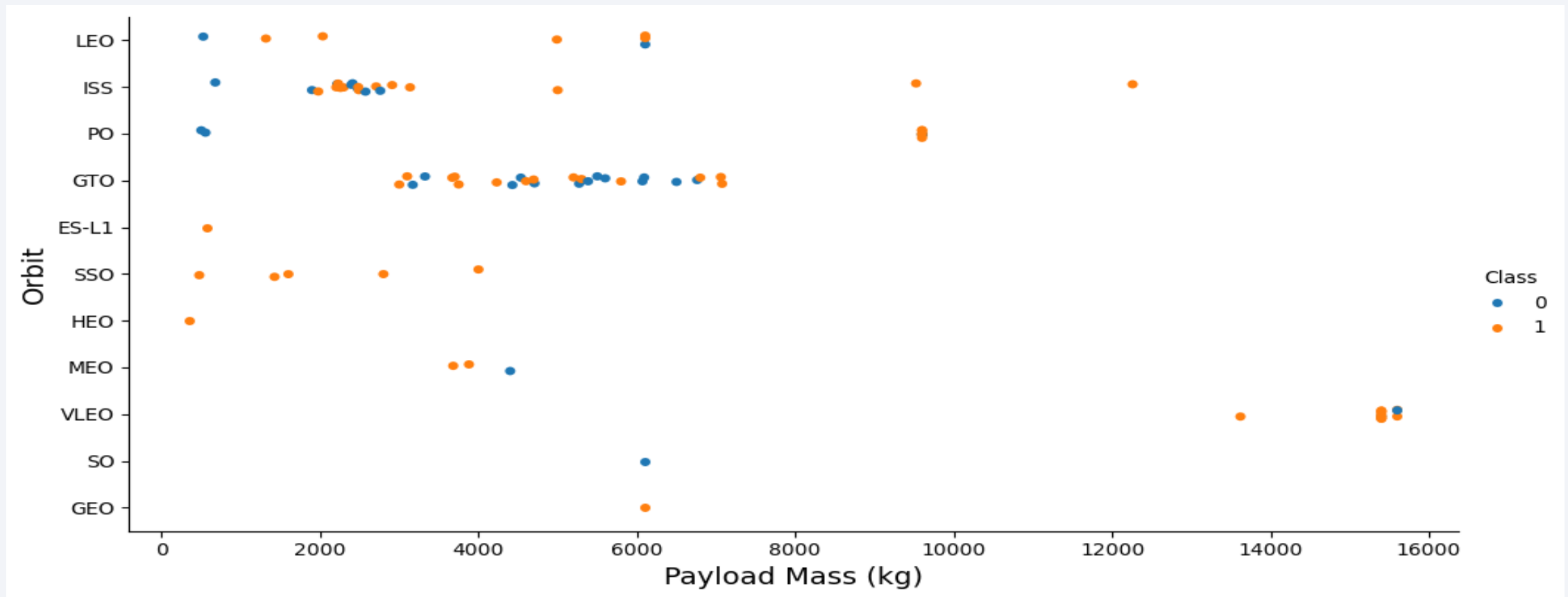
- Each bar shows the average success rate for missions to different orbits. **SSO, ES-L1, GEO and HEO** have the highest success, while **GTO, ISS, PO and LEO** have lower rates.
- Orbit with 0% success rate is **SO**.

Flight Number vs. Orbit Type



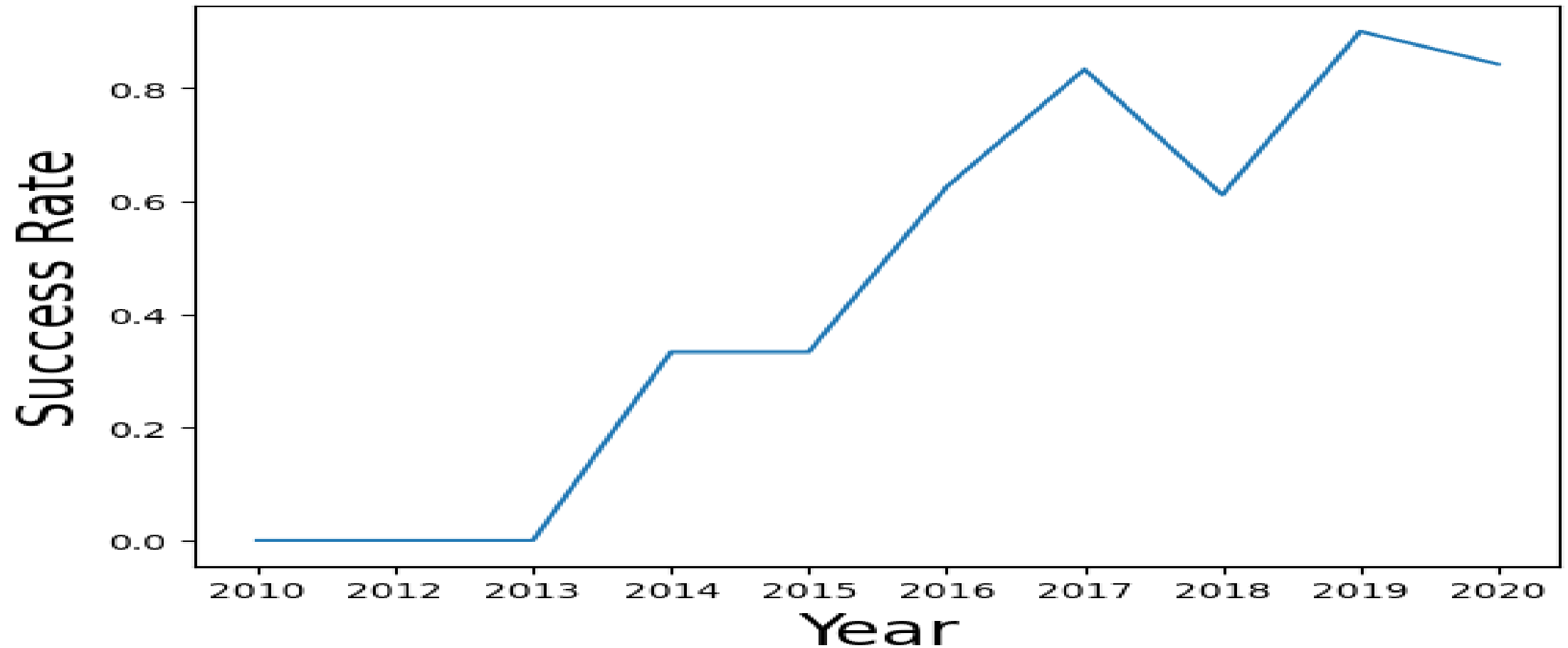
This scatter plot shows how SpaceX flight numbers relate to orbit types, with color indicating mission success (orange) or failure (blue). It reveals that as flight numbers increase, success becomes more common across most orbits - especially LEO and SSO - highlighting SpaceX's growing reliability over time.

Payload vs. Orbit Type



Mission success seems more likely in **SSO** and **LEO**, especially with moderate payloads. **GTO** missions show more failures, possibly due to higher complexity or tighter launch constraints.

Launch Success Yearly Trend



- The line chart groups the data by year, computes the average success (where 1 = success, 0 = failure), and visualizes the trend using a line chart. The result shows a clear upward trajectory in success rate from 2013 to 2020.

All Launch Site Names

```
%sql SELECT DISTINCT "Launch_Site" FROM SPACEXTABLE;
```

Done.

```
Launch_Site  
CCAFS LC-40  
VAFB SLC-4E  
KSC LC-39A  
CCAFS SLC-40
```

Displaying the names of the unique launch sites in the space mission.

Launch Site Names Begin with 'CCA'

```
%sql SELECT * FROM SPACEXTABLE WHERE "Launch_Site" LIKE 'CCA%' LIMIT 5;
```

Done.

```
(5 rows: Date, Time, Booster_Version, Launch_Site='CCAFS LC-40',  
Payload, Payload_Mass, Orbit, Customer, Outcome, Landing_Outcome)
```

This is a SQL query of 5 records where launch sites names begin with CCA.

Total Payload Mass

```
%sql SELECT SUM(PAYLOAD_MASS_KG_) AS Total_Payload_Mass  
FROM SPACEXTABLE WHERE Customer = 'NASA (CRS)';
```

Done.

Total_Payload_Mass
45596

This SQL query calculates the total payload carried by boosters from NASA.

Average Payload Mass by F9 v1.1

```
%sql SELECT AVG(PAYLOAD_MASS_KG_) AS Average_Payload_Mass  
FROM SPACEXTABLE WHERE Booster_Version = 'F9 v1.1';
```

Done.

Average_Payload_Mass
2928.4

This SQL query calculates the Average payload mass carried by booster version F9 v1.1.

First Successful Ground Landing Date

```
%sql SELECT MIN(Date) AS First_Successful_Ground_Landing  
FROM SPACEXTABLE WHERE Landing_Outcome = 'Success (ground pad)';
```

Done.

```
First_Successful_Ground_Landing  
2015-12-22
```

Dates of the first successful landing outcome on ground pad.

Successful Drone Ship Landing with Payload between 4000 and 6000

```
%sql SELECT Booster_Version FROM SPACEXTABLE  
WHERE Landing_Outcome = 'Success (drone ship)'  
AND PAYLOAD_MASS_KG_ > 4000 AND PAYLOAD_MASS_KG_ < 6000;
```

Done.

```
Booster_Version  
F9 FT B1022  
F9 FT B1026  
F9 FT B1021.2  
F9 FT B1031.2
```

Names of boosters which have successfully landed on drone ship and had payload mass greater than 4000 but less than 6000.

Total Number of Successful and Failure Mission Outcomes

```
%sql SELECT Mission_Outcome, COUNT(*) AS Total  
FROM SPACEXTABLE GROUP BY Mission_Outcome;
```

Done.

Mission_Outcome	Total
Failure (in flight)	1
Success	98
Success	1
Success (payload status unclear)	1

This SQL query counts SpaceX missions by their outcome type. Shows 98 clear successes, 1 in-flight failure, and 2 ambiguous cases(success-1, success(payload status unclear)).

Boosters Carried Maximum Payload

```
%sql SELECT DISTINCT Booster_Version FROM SPACEXTABLE  
WHERE PAYLOAD_MASS_KG_ = (SELECT MAX(PAYLOAD_MASS_KG_) FROM SPACEXTABLE);
```

Done.

Booster_Version

F9 B5 B1048.4	F9 B5 B1049.4	F9 B5 B1051.3
F9 B5 B1056.4	F9 B5 B1048.5	F9 B5 B1051.4
F9 B5 B1049.5	F9 B5 B1060.2	F9 B5 B1058.3
F9 B5 B1051.6	F9 B5 B1060.3	F9 B5 B1049.7

Names of the different boosters which carried the maximum payload mass.

2015 Launch Records

```
%sql SELECT substr(Date,6,2) Month, Landing_Outcome, Booster_Version, Launch_Site
FROM SPACEXTABLE WHERE substr(Date,0,5)='2015' AND Landing_Outcome='Failure (drone ship)';
```

Done.

Month	Landing_Outcome	Booster_Version	Launch_Site
01	Failure (drone ship)	F9 v1.1 B1012	CCAFS LC-40
04	Failure (drone ship)	F9 v1.1 B1015	CCAFS LC-40

- This SQL query task filters SpaceX launches from 2015 that failed to land on drone ships. It extracts the month from the date and displays the landing outcome, booster version, and launch site. Only two such failures occurred—in January and April—both using F9 v1.1 boosters at CCAFS LC-40.

Rank Landing Outcomes Between 2010-06-04 and 2017-03-20

```
%sql SELECT Landing_Outcome, COUNT(*) Outcome_Count FROM SPACEXTABLE
WHERE Date BETWEEN '2010-06-04' AND '2017-03-20'
GROUP BY Landing_Outcome ORDER BY Outcome_Count DESC;
```

Done.

Landing_Outcome	Outcome_Count
No attempt	10
Success (drone ship)	5
Failure (drone ship)	5
Success (ground pad)	3
Controlled (ocean)	3
Uncontrolled (ocean)	2
Failure (parachute)	2
Precluded (drone ship)	1

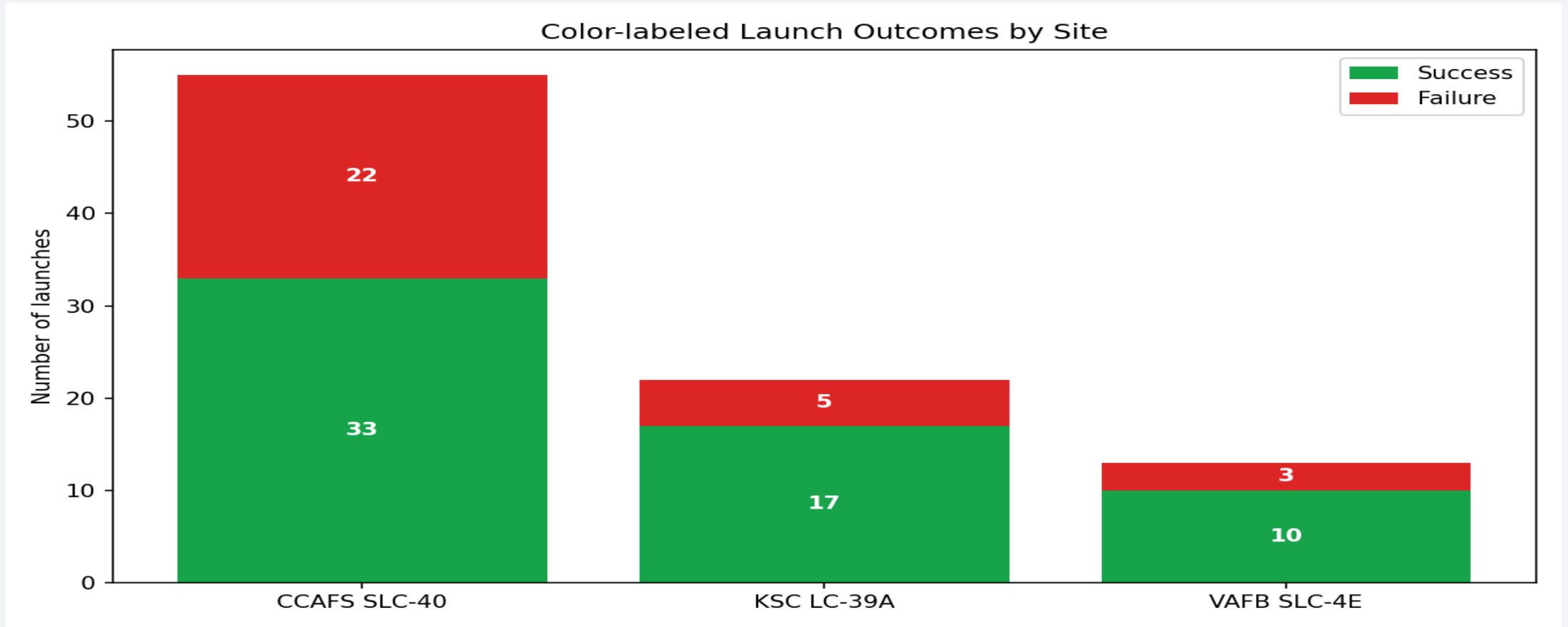
- Between 2010 and 2017, SpaceX mostly did not attempt landings, focusing on launch reliability. Drone ship landings had mixed results - 5 successes and 5 failures - while ground pad landings were more successful. This period marked their transition from testing to mastering reusable rocket technology.

A satellite view of Earth from space, showing the curvature of the planet and city lights at night. The image is a composite of a dark blue sky and a view of the Earth's surface, which is covered in clouds and illuminated by city lights. The text "Section 3" is overlaid on the left side of the image.

Section 3

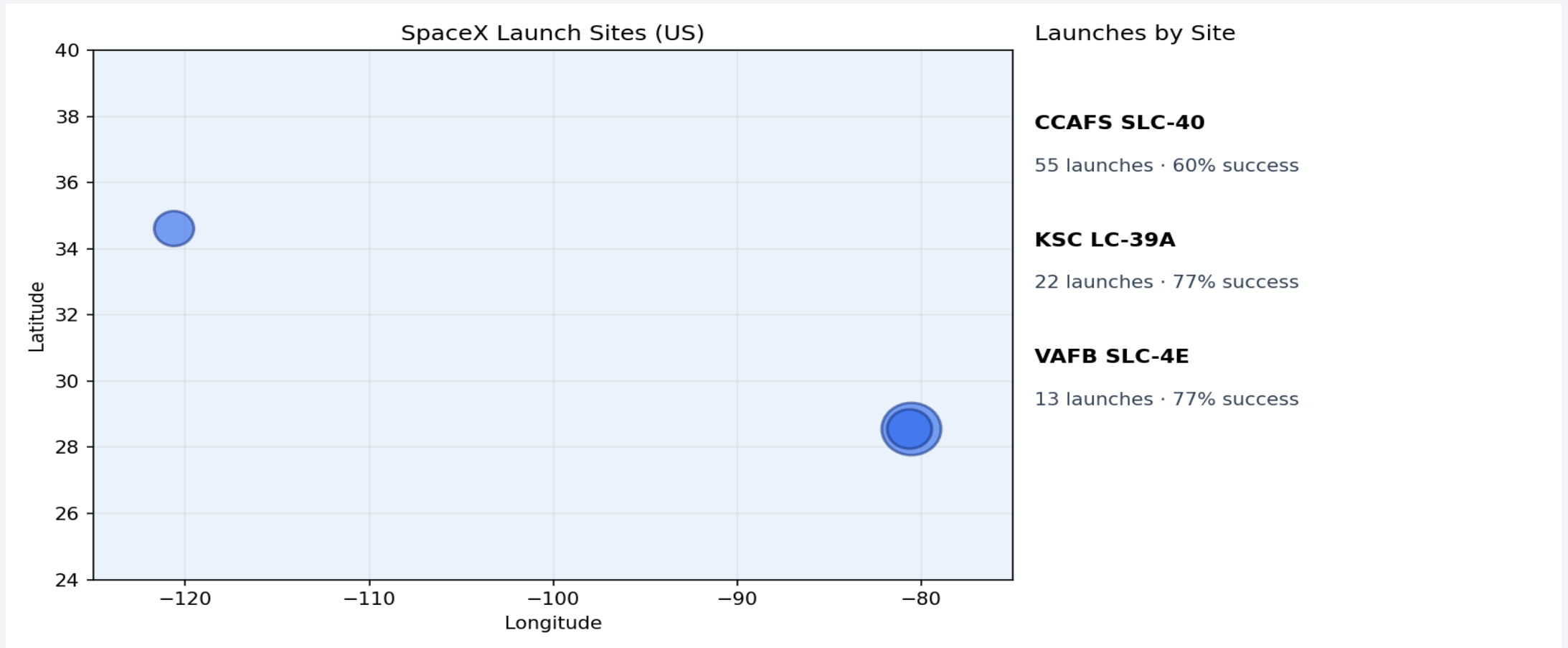
Launch Sites Proximities Analysis

Color labeled launch outcomes

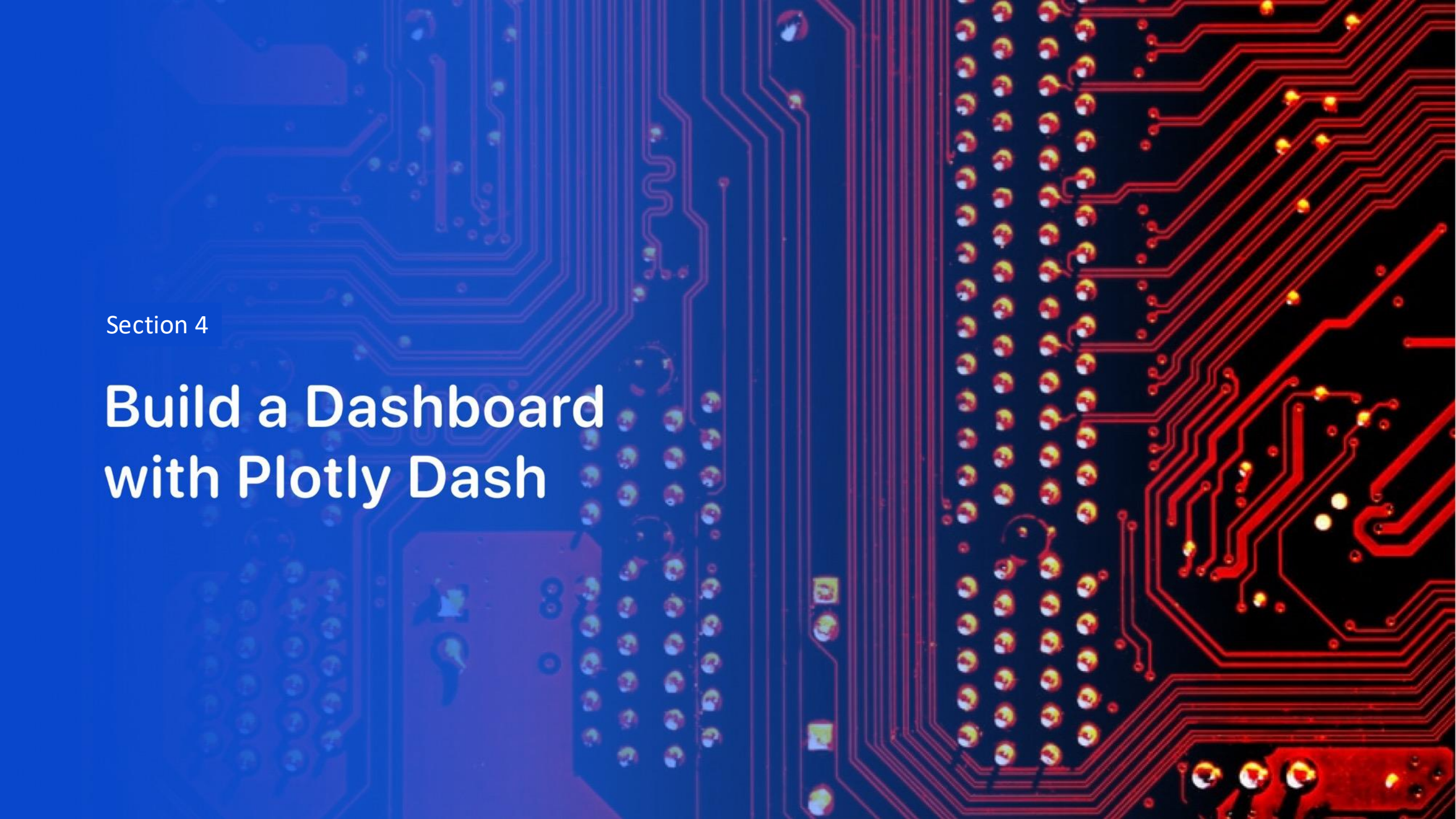


This chart summarizes the color-labeled launch outcomes shown on the interactive folium map, with green markers for successful landings and red for

Folium Map



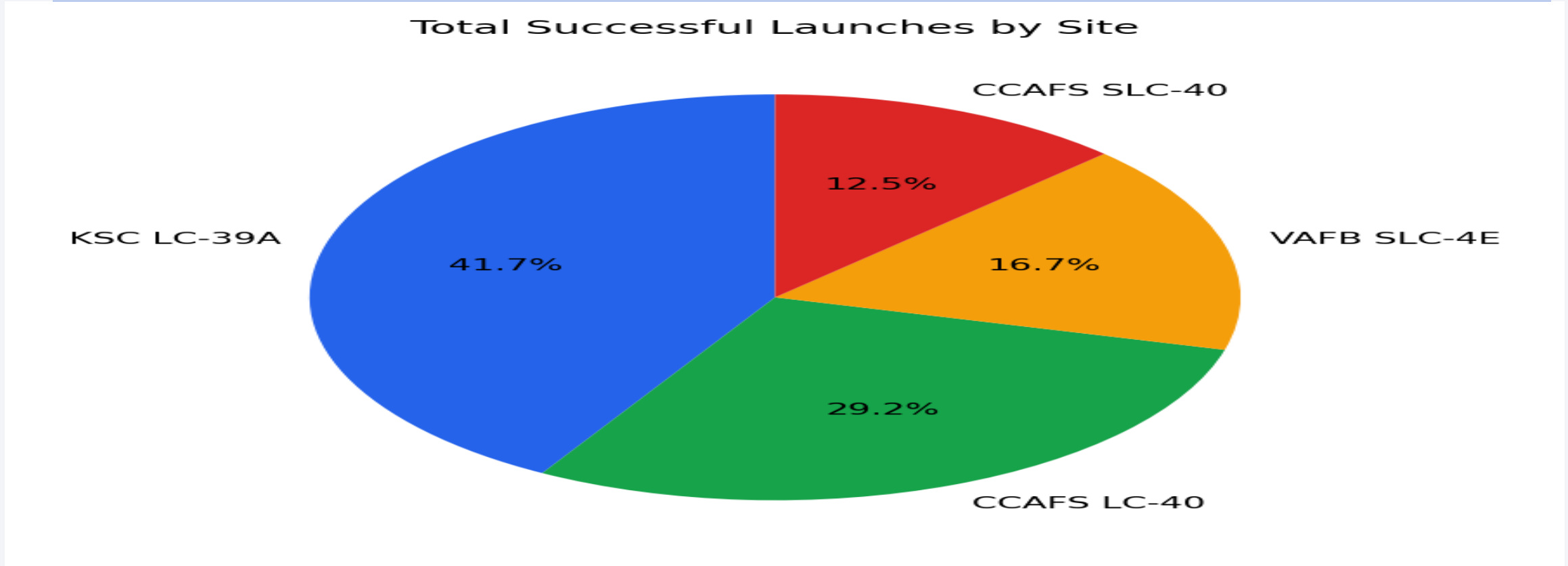
This folium map shows the SpaceX launch sites in the U.S.: **VAFB SLC-4E** in California, and two sites at Cape Canaveral / KSC in **Florida**. Each marker displays the number of launches from that site, with success rate shown alongside



Section 4

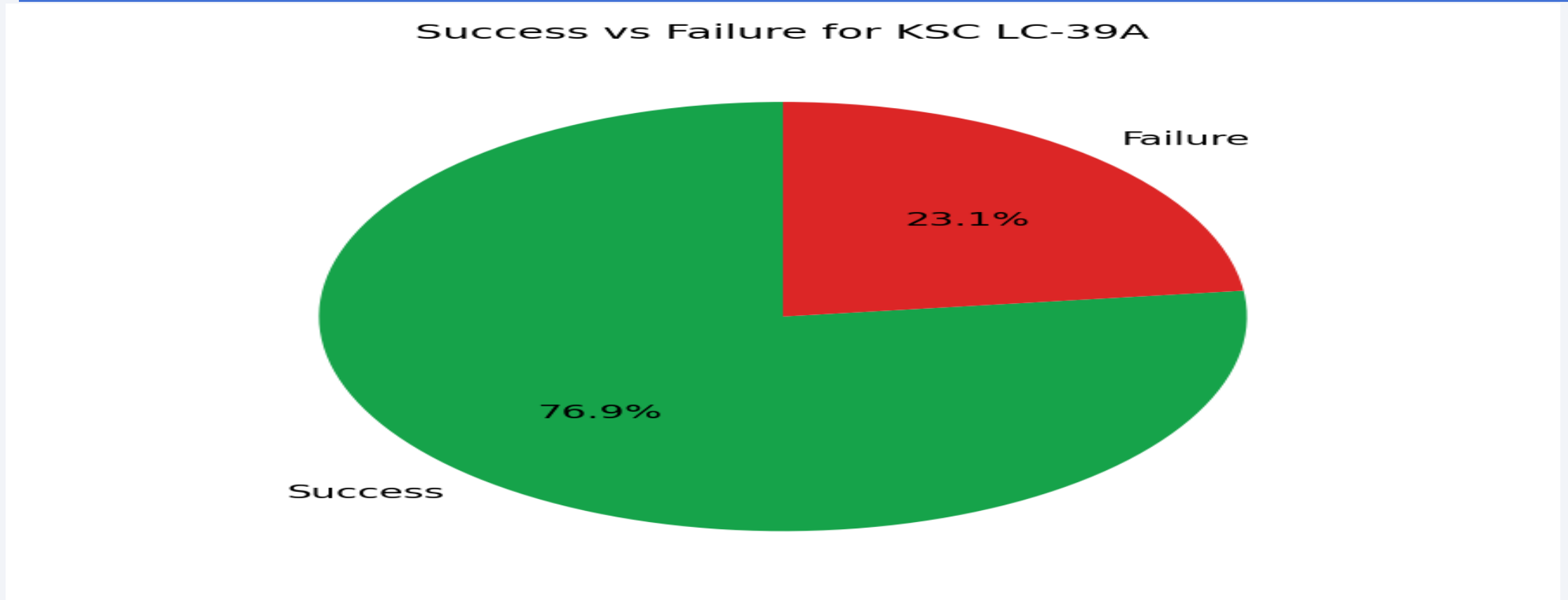
Build a Dashboard with Plotly Dash

Launch success counts for all sites



- KSC LC-39A leads with 41.7% of all successful launches. This site is clearly SpaceX's most reliable or frequently used for successful missions. KSC LC-39A is likely the preferred site for high-profile or heavy-lift missions, given its strong success record.
- CCAFS SLC-40 trails with 12.5%, suggesting it's either less active or has a lower success rate.

Launch Site with better failure vs success ratio

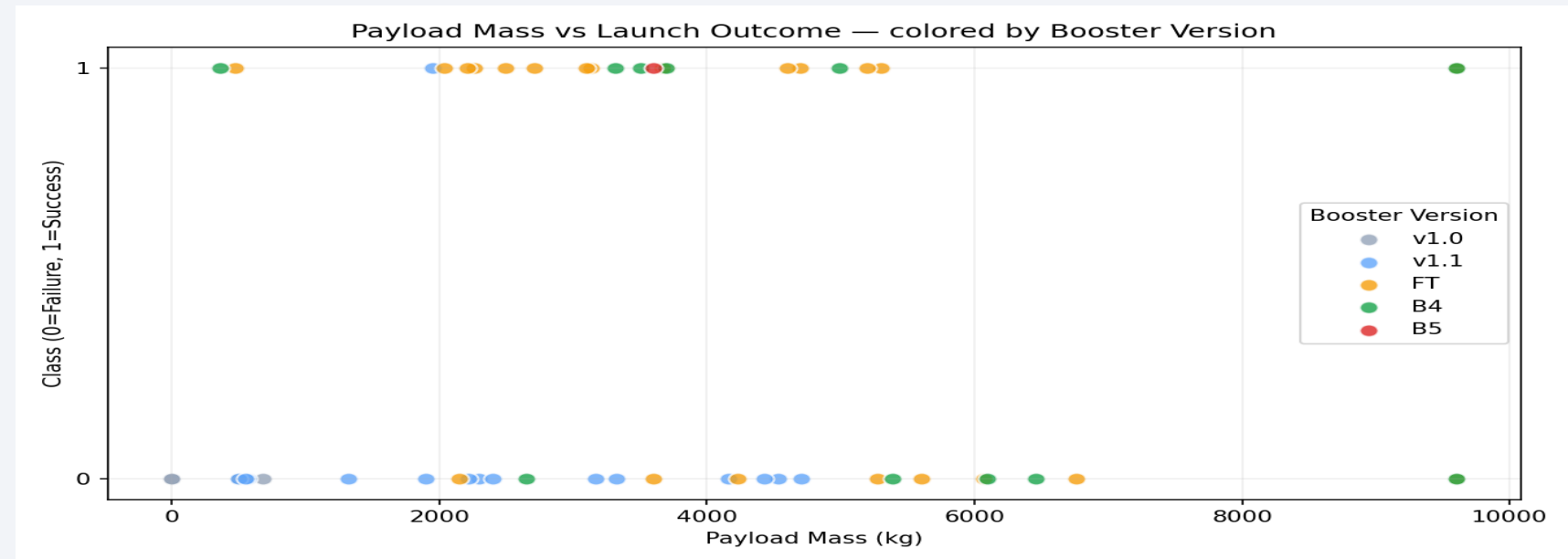
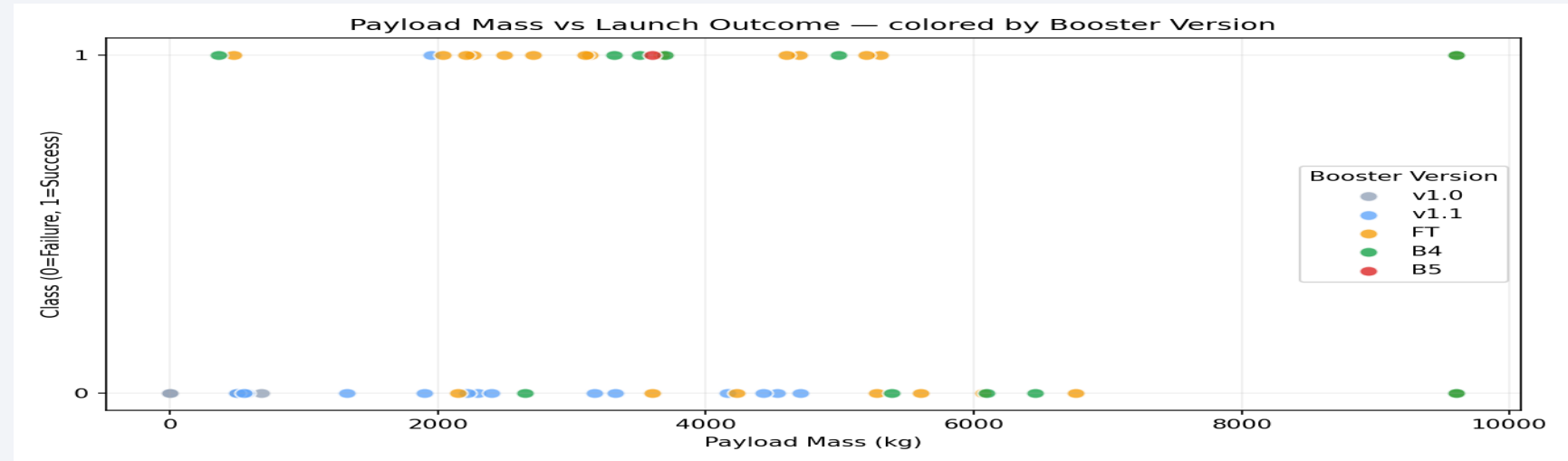


- Across all launch sites, a consistent pattern emerges: failure rates exceed success rates. Yet, this particular site stands out with a comparatively lower failure rate—an unexpected finding considering SpaceX's broader reputation for high reliability.

Payload Mass vs Launch Outcome(all sites)

Explanation:

From the chart, I observed that payloads between 2034kg and 5000kg have the highest success rate.

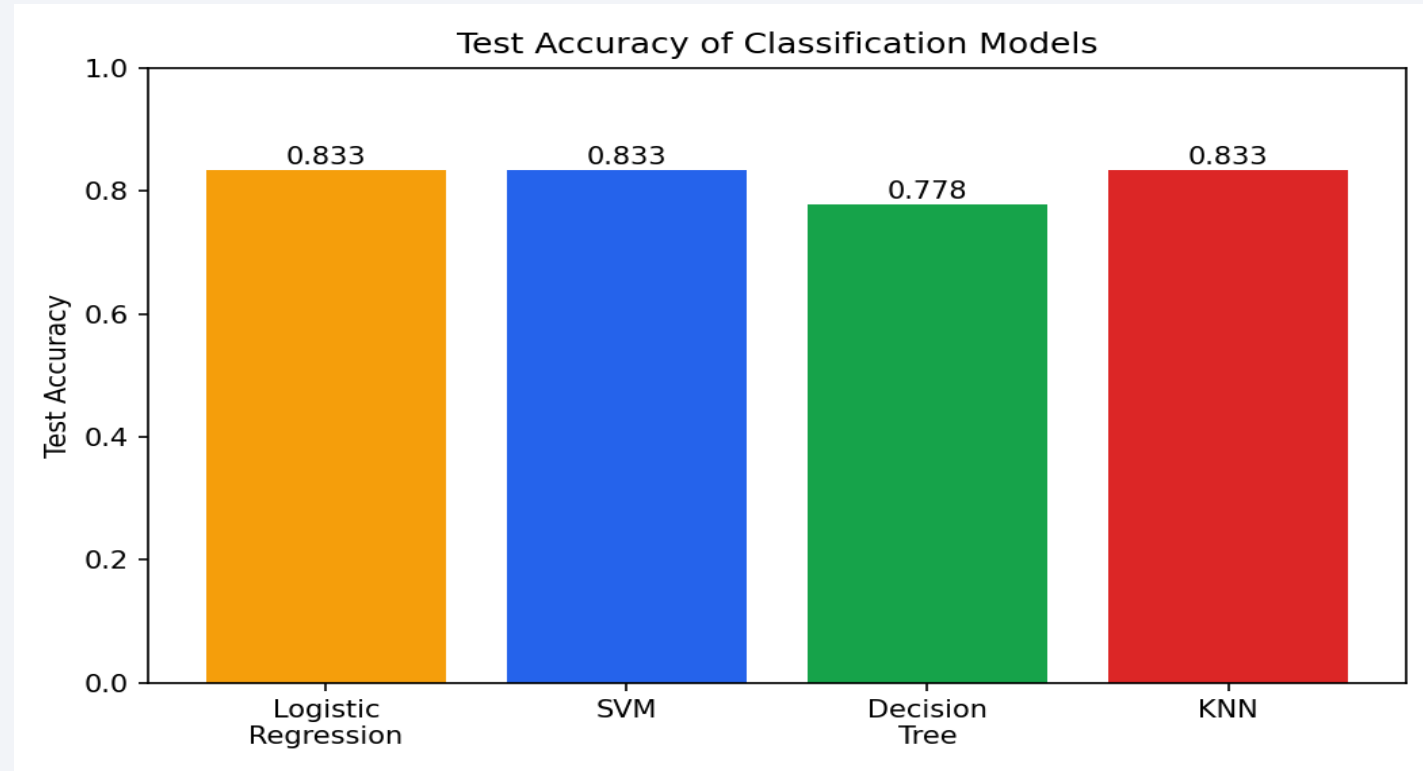


Section 5

Predictive Analysis (Classification)

Classification Accuracy

- Decision Tree had the lowest test accuracy (0.778) of the four models, while Logistic Regression (CV 0.846) and KNN (CV 0.848) had the strongest cross-validated scores.
- Logistic Regression, SVM and KNN all show equally good performance.
- However, I would pick Logistic regression because:
- Simple model, easiest to interpret and low computational cost .



Model	Best CV Accuracy	Test Accuracy
Logistic Regression	0.8464	0.8333
SVM	n/a	0.8333
KNN	0.8482	0.8333
Decision Tree	n/a	0.7778

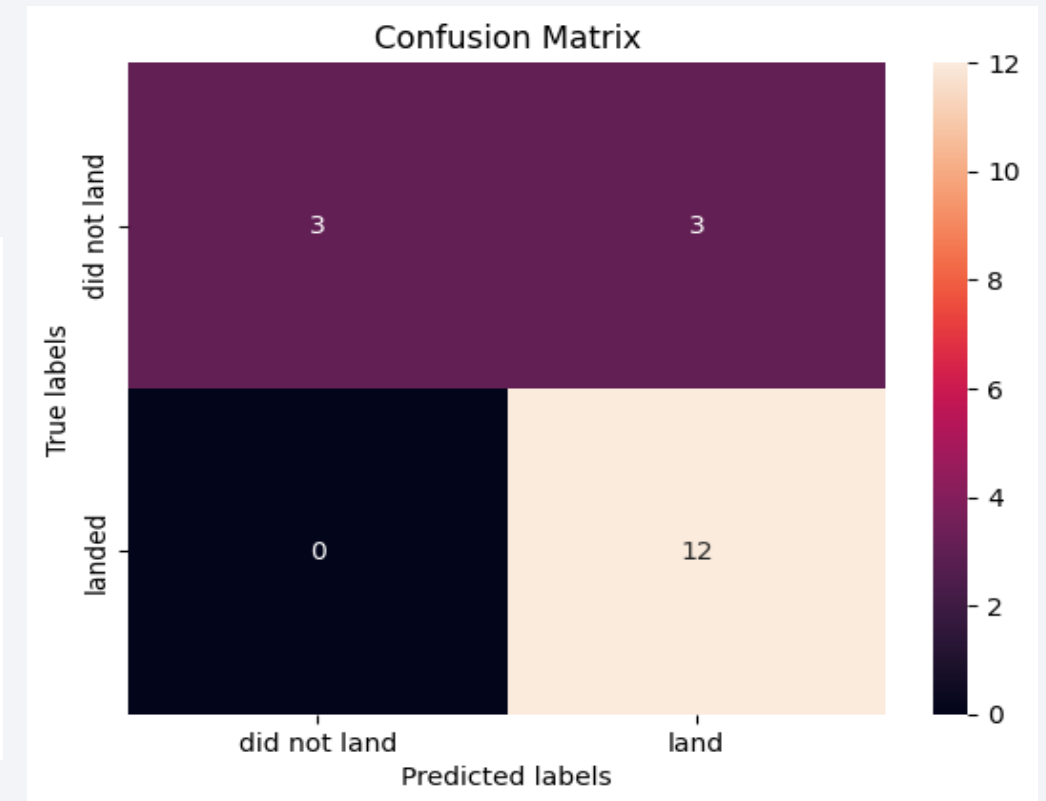
Confusion Matrix

- Explanation:

Based on the confusion matrix, we see that logistic regression can distinguish between the different classes.

- True Positive(TP) - 12 (True label is landed, Predicted label is also landed)
- False Positive(FP) - 3 (True label is not landed, Predicted label is landed)
- The model is very good at detecting landings (Recall = 1.0) it never misses a successful landing.
- However, it makes some false positives (predicting “landed” when the rocket actually didn’t land).
- This means the model is optimistic about landings, favoring recall over precision.

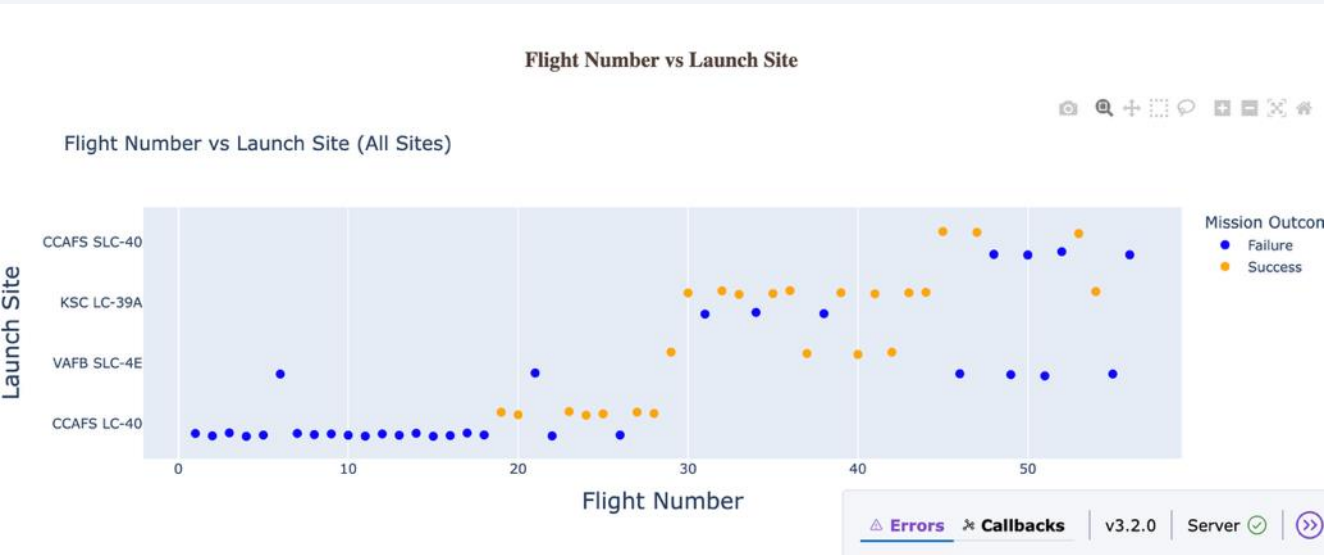
	NEGATIVE	POSITIVE
NEGATIVE	Count of TN	Count of FP
POSITIVE	Count of FN	Count of TP



Conclusions

- Logistic Regression, SVM and KNN all showed equally good performance.
- KSC LC-39A leads with 41.7% of all successful launches. This site is clearly SpaceX's most reliable or frequently used for successful missions. KSC LC-39A is likely the preferred site for high-profile or heavy-lift missions, given its strong success record.
- CCAFS SLC-40 trails with 12.5%, suggesting it's either less active or has a lower success rate.
- Decision Tree had the lowest test accuracy (0.778) of the four models tested.

Appendix



```
# Assign marker colors based on the 'class' column
spacex_df['marker_color'] = spacex_df['class'].apply(lambda x: 'green' if x == 1 else 'red')

TODO: For each launch result in spacex_df data frame, add a folium.Marker to marker_cluster

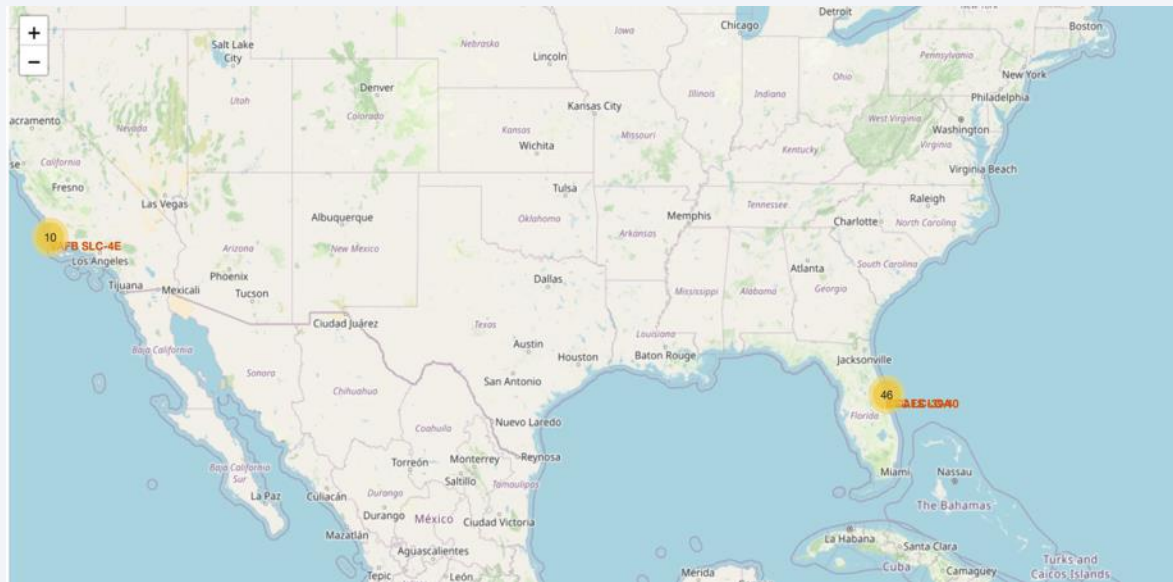
17]: # Add marker_cluster to current site_map
site_map.add_child(marker_cluster)

# for each row in spacex_df data frame
# create a Marker object with its coordinate
# and customize the Marker's icon property to indicate if this launch was succeeded or failed,
# e.g., icon=folium.Icon(color='white', icon_color=row['marker_color'])
for index, record in spacex_df.iterrows():

    # TODO: Create and add a Marker cluster to the site map
    lat = record['Lat']
    lon = record['Long']

    marker = folium.Marker(
        location=[lat, lon],
        icon=folium.Icon(color='white', icon_color=record['marker_color']),
        popup=f"Launch Site: {record['Launch Site']}<br>Outcome: {'Success' if record['class'] == 1 else 'Failure'}"
    )
    marker_cluster.add_child(marker)

site_map
```



Thank you!

